

§1

$$\frac{d}{dx} \int_0^{g(x)} f(t) dt = f(g(x)) g'(x).$$

~~$$\hat{a}|0\rangle = 0.$$~~

~~$$\psi = \frac{p}{2m} + \frac{1}{2} m \omega^2 x^2$$~~

$$\hat{a}|n\rangle = \sqrt{n} |n-1\rangle$$

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1} |n+1\rangle$$

§2

M1

$$\lim_{x \rightarrow 0} \frac{\int_0^x (t - \sin t) dt}{x^2(e^x - 1)} = \lim_{x \rightarrow 0} \frac{x - \sin x}{2x^2 + 3x^3 + 2x^4}$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^2 + \cos x - 1}{x^2(e^x - 1)} = \lim_{x \rightarrow 0} \frac{\frac{1}{2}x^2 + 1 - \frac{1}{2}x^2 + \frac{x^4}{24} + O(x^6) - 1}{x^2(\frac{x}{1} + O(x^2))} = \frac{1}{24}$$

Q1

①

$$\hat{a} \psi_0(x) = 0.$$

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + \frac{i}{\sqrt{2m\hbar\omega}} \hat{p}$$

$$\sqrt{\frac{m\omega}{2\hbar}} x \psi_0 + \frac{i}{\sqrt{2m\hbar\omega}} -i\hbar \frac{d}{dx} \psi_0 = 0$$

$$\sqrt{\frac{m\omega}{2\hbar}} x \psi_0 = -\frac{\hbar}{\sqrt{2m\hbar\omega}} \frac{d}{dx} \psi_0$$

$$\frac{1}{\psi_0} d\psi_0 = -\frac{m\omega}{\hbar} x dx$$

$$\psi_0 = A e^{-\frac{m\omega}{2\hbar} x^2}$$

$$\int |\psi_0|^2 dx = \int A^2 e^{-\frac{m\omega}{\hbar} x^2} dx = A^2 \int e^{-\frac{m\omega}{\hbar} x^2} dx = A^2 \sqrt{\frac{\hbar}{m\omega}}$$

$$A = \left(\frac{m\omega}{\hbar\pi}\right)^{\frac{1}{4}}$$

$$\psi_0(x) = \left(\frac{m\omega}{\hbar\pi}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar} x^2}$$

(2)

$$\begin{cases} \hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) \\ \hat{p} = \frac{i\hbar m\omega}{2} (\hat{a} - \hat{a}^\dagger) \end{cases}$$

$$\langle m | \hat{x} | n \rangle = \langle m | \sqrt{\frac{\hbar}{2m\omega}} (\hat{a} + \hat{a}^\dagger) | n \rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \langle m | \hat{a} | n \rangle + \sqrt{\frac{\hbar}{2m\omega}} \langle m | \hat{a}^\dagger | n \rangle$$

$$= \sqrt{\frac{n\hbar}{2m\omega}} \langle m | n-1 \rangle + \sqrt{\frac{(n+1)\hbar}{2m\omega}} \langle m | n+1 \rangle$$

Ex: $m=n-1$ $\langle m | \hat{x} | n \rangle = \sqrt{\frac{n\hbar}{2m\omega}}$
 $m=n+1$ $\langle m | \hat{x} | n \rangle = \sqrt{\frac{(n+1)\hbar}{2m\omega}}$

(3)

$$\langle n | \hat{x} | n \rangle = 0 \quad \langle n | \hat{p} | n \rangle = 0$$

$$(\delta x)^2 = \langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2$$

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$$= \langle \hat{x}^2 \rangle$$

$$= \langle n | \frac{\hbar}{2m\omega} (\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}) | n \rangle$$

$$[\hat{a}, \hat{a}^\dagger] = 1 \Rightarrow \langle n | \frac{\hbar}{2m\omega} \langle n | 2\hat{a}\hat{a}^\dagger | n \rangle$$

$$= \frac{\hbar}{2m\omega} (2n+1)$$

$$(\delta p)^2 = \langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2$$

$$= \langle n | \frac{m\hbar\omega}{2} (\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}) | n \rangle$$

$$= \frac{m\hbar\omega}{2} \langle n | \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} | n \rangle$$

$$= \frac{m\hbar\omega}{2} (2n+1)$$

$$(\delta x)(\delta p) = \frac{\hbar}{4} (2n+1)^2 \quad \text{for } n=0 \quad \delta x \delta p \geq \frac{\hbar}{2}$$